

Spatial and Temporal Entropy for IoT Fault Detection: A Comparative Study with Generative References

Alexander V. Mantzaris and James H. Korris

Abstract—Spatial covariance and within-sensor temporal order describe different aspects of IoT behavior. We compare local correlation-spectrum, permutation and sample entropy with conventional spatial and temporal statistics under joint block-bootstrap and graph-conditioned diffusion references. A frozen extension evaluates 360 fault realizations on 20 simulation/recording blocks with three model seeds, using one synthetic benchmark and two semi-synthetic real-data benchmarks. Aligned diagnostics verify that time reordering can preserve covariance while changing temporal entropy; autocorrelation also detects this distinction. Results depend on support, reference and calibration. On Intel, bootstrap temporal entropy achieves recall 0.389 versus 0.000 for the conventional temporal scan, with respective background exceedance 0.103 and 0.000. Adding entropy to the conventional combined scan increases synthetic bootstrap recall by 0.003; other dataset/reference combinations show no recall increase. Shared-support fidelity exposes substantial diffusion entropy miscoverage. No consistent overall advantage is established for either entropy or the generative reference. These exploratory results distinguish feature information from estimator availability and decision performance, without claiming equivalence.

Index Terms—IoT, spatial entropy, permutation entropy, sample entropy, diffusion forecasting, fault localization

I. INTRODUCTION

Sensor-fed digital twins and data spaces need evidence about changes in both collective organization and individual dynamics. Copying streams can reduce independent spatial variation; replay can alter temporal organization without creating a large marginal outlier. A legitimate operating transition can also change both. An alert should therefore expose measurements and their provenance, rather than assert an unverified physical cause.

This study compares information captured by spatial and temporal entropy with conventional measurements, separating the feature from its normal-reference model. The motivating connection to Mantzaris’s work is specific: entropy trajectories describe evolving segregation under different network topologies [1] and collective flocking organization [2]. We transfer attention to evolving organization and interaction structure. The segregation study uses degree/homophily distributions; the flocking work develops a thermodynamic description of its simulation. Here spatial entropy uses normalized correlation eigenvalues, permutation entropy uses ordinal patterns, and sample entropy uses matching templates. These definitions differ. We infer neither thermodynamic quantities nor a second law for sensor recordings.

The earlier version of this project evaluated spatial correlation-spectrum entropy and established no consistent improvement over its synchronization comparators. That finding motivates, but does not predetermine, this extension. A changing spatial-entropy trace is not itself a within-sensor temporal-entropy estimator. Nor does an unsuccessful reference model establish a limitation of all entropy features. The original artifacts remain identifiable; new comparisons, support corrections and diagnostics have separate manifests.

We address six questions. **RQ1**: does spatial entropy improve detection or localization over non-entropic spatial statistics? **RQ2**: do temporal entropies reveal changes missed by spatial covariance summaries? **RQ3**: do they improve over conventional temporal statistics? **RQ4**: does combining spatial and temporal evidence improve detection or localization? **RQ5**: does a generative reference improve over joint block bootstrap? **RQ6**: how do support, missingness, window length, calibration and mechanism affect these answers?

Our contribution is a reproducible comparison, elementary theoretical characterization of the information retained by its measurements, and an inspectable monitoring artifact. We introduce neither an entropy definition nor a general-purpose diffusion architecture. Conventional features augmented with entropy are compared directly with conventional features alone; entropy is not required to win or lose.

II. RELATED WORK

Effective rank measures spectral dimensionality through entropy of normalized singular values [3]; these equal eigenvalues for a PSD correlation matrix. Graph-entropy anomaly monitoring also predates this study [4], although a graph Laplacian spectrum differs from the correlation spectrum used here. Spatial meaning comes from physical/proximity groups, not from the ordering of sensor columns.

Bandt and Pompe’s permutation entropy summarizes neighboring ordinal patterns [5]. Richman and Moorman’s sample entropy describes template-extension regularity [6]. Their probability objects, support requirements and amplitude sensitivity differ. Costa et al.’s multiscale sample entropy [7] explicitly warns against identifying irregularity with healthy complexity. We separately label coarse-grained permutation and sample entropy and make no physiological-health inference.

TimeGrad [8], CSDI [9] and DiffSTG [10] establish diffusion forecasting, conditional imputation and graph-conditioned generation. DiffAD [11], ImDiffusion [12] and ICDiffAD [13]

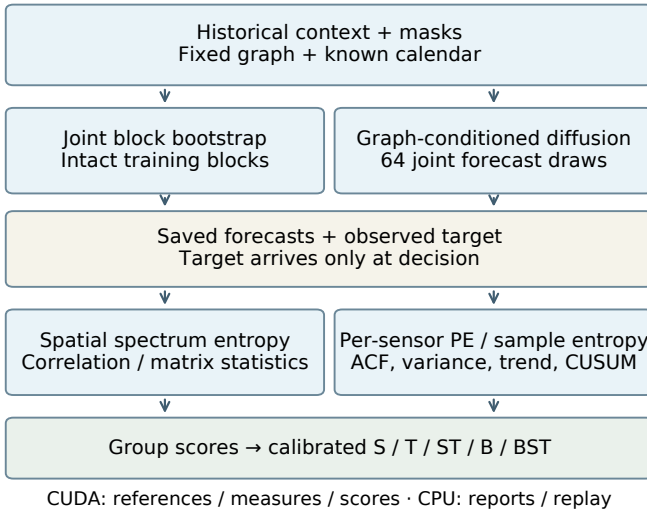


Fig. 1. Feature family and normal reference are separate experimental factors. Both generators issue a joint trajectory before the target; support is applied after generation. Calibration includes the complete frozen scan family.

already apply diffusion to anomaly detection. Our generator uses a forecasting boundary: it receives no measured target value. GDN [14] supplies a graph-prediction comparator, with adaptation and information access disclosed below. Dependent-data conformal work [15] motivates explicit calibration assumptions; we do not claim a new distribution-free guarantee for an indefinite sensor stream.

III. MEASUREMENTS, REFERENCES AND DECISIONS

A. Groups, channels and spatial measurements

Fixed graph-hop neighborhoods have sizes $m \in \{6, 12, 24\}$ around 24 index-spaced centers, deduplicated within size. Geographic distance breaks finite hop ties; node order resolves unreachable nodes when a component is too small. Graphs represent actual synthetic coupling, Intel coordinate proximity, or release road distances. Channels remain separate. Main extension windows are $W \in \{48, 96\}$ with $d = W/4$; the original $W = 24$ scan is preserved separately.

For one group and channel, retain common complete rows, requiring $n \geq \max(2m, \lceil .8W \rceil)$. If any column has sample variance at most 10^{-8} in training-scaled units, that group/channel correlation is undefined and quality-flagged. Otherwise standardize all m columns to sample variance one, giving Z :

$$\mathbf{R}_{g,t} = Z^\top Z / (n - 1), \quad (1)$$

$$\tilde{\mathbf{R}}_{g,t} = (1 - \lambda)\mathbf{R}_{g,t} + \lambda I, \quad (2)$$

$$p_k = \lambda_k(\tilde{\mathbf{R}}_{g,t}) / \text{tr}(\tilde{\mathbf{R}}_{g,t}), \quad (3)$$

$$H_S(g, t) = - \sum_k p_k \log p_k / \log m, \quad (4)$$

$$\Delta H_S(g, t) = H_S(g, t) - H_S(g, t - d). \quad (5)$$

Here $0 \log 0 = 0$ and $\lambda = .05$. The symmetric eigensolver rejects materially non-PSD matrices; only roundoff within $64m$ machine epsilons is tolerated. Gram products disable

TF32. This is *local correlation-spectrum entropy*, not entropy of coordinates, thermodynamic states or a degree distribution.

Spatial conventional features are mean signed correlation C , mean absolute correlation, standardized pairwise disagreement D , leading concentration $\lambda_{\max}(\mathbf{R})/m$, and their lagged changes. With sample-standardized columns, $D = 2(n - 1)(1 - C)/n$, so it is redundant with C at fixed support. Full-matrix comparisons include reference-standardized Frobenius discrepancies of both current $\mathbf{R}$ and $\Delta \mathbf{R}$. A scalar function of $\mathbf{R}$ cannot contain more information than the full matrix; it may still differ statistically in estimation or calibration.

B. Per-sensor temporal entropy and support

We keep each sensor's measurements before group aggregation. For sensor i , use the ordinal vector $v_{i,s} = (x_{i,s+j\tau})_{j=0}^{q-1}$ and estimate its pattern frequencies $p_{i,t}(\pi)$ in the current window:

$$H_{PE}(i, t) = - \sum_{\pi} p_{i,t}(\pi) \log p_{i,t}(\pi) / \log(q!). \quad (6)$$

The initial configuration is $q = 3, \tau = 1$; development compares $(3, 1), (3, 2), (4, 1)$. Missing timestamps invalidate affected templates without compressing time. Stable temporal-index tie breaking is deterministic. Both endpoints require at least $\max(30, 5q!)$ valid templates; a tied-template fraction above .5 or a flatline produces an entropy abstention. Tied fractions and a drop-ties development sensitivity are retained. Ordinal diversity is invariant to a strict monotone amplitude transformation (up to relabeling patterns); high entropy does not automatically mean healthy complexity.

For sample entropy, construct complete $(r + 1)$ -templates with $r = 2$. The common admissible pair set contains unordered starts $a < b$ with $b - a > 2$, excluding self matches and a predeclared Theiler neighborhood. Let D_r be the maximum distance in the first r coordinates, and D_{r+1} include the successor. Then

$$B_r = \sum_{(a,b) \in \mathcal{P}} \mathbf{1}\{D_r(a, b) \leq \delta\}, \quad (7)$$

$$A_{r+1} = \sum_{(a,b) \in \mathcal{P}} \mathbf{1}\{D_{r+1}(a, b) \leq \delta\}, \quad (8)$$

$$H_{SE}(i, t) = - \log(A_{r+1}/B_r). \quad (9)$$

Both counts use exactly the same valid pairs. Tolerance candidates .1/.2/.3 multiply the sensor's fixed training scale, not its current variance. Development selects finite-estimate support, with declared tie preferences. We retain A, B , pair and template counts. $B = 0$ is undefined; $A = 0 < B$ is censored/infinite. Operational scores require 30 templates, $B \geq 20, A > 0$, nonconstant data and valid endpoints. No pseudocount or arbitrary large finite value replaces an undefined estimate. This conservative scoring rule can reduce availability; raw support and censoring remain visible.

For both entropies we compute level and lagged change. Limited multiscale diagnostics average nonoverlapping bins of sizes 1, 2 and 4 aligned to the left window boundary. Every bin member must be observed; incomplete final bins are discarded.

The same template criteria apply after coarsening. We do not enlarge the forecast horizon merely to rescue an unsupported scale. Conventional comparators receive identical history in these diagnostics. A separately frozen exploratory sensitivity compares scales 1 and 2 on W96 with the same physical $d = 24$, seed-17 episodes and references. It refits scale-2 floors on the original four development issuances and recalibrates each family; other selected settings are fixed. PE/ACF/Theiler lags use the coarse grid, so their physical delays double. Spatial measurements retain the original grid.

C. Conventional temporal measurements and feature families

Temporal baselines include ACF at lags 1, 2 and 4, changes in that ACF vector, variance of first differences, local sample variance, least-squares trend, predictive deviation of local mean, and a two-sided within-window CUSUM. Lag pairs use actual grid positions. Each statistic requires at least $\max(12, \lceil W/2 \rceil)$ points or applicable pairs. CUSUM uses its final state with allowance .5 in training-scaled units, resets at the window start and pauses on missing rows. Reference scoring below makes these measurements conditional deviations. Raw predictive residual and the original GDN adaptation remain additional detectors. Raw residuals average pointwise absolute deviations divided by generated MAD plus .05, and retain the original spatial-quality restriction. The labeled GDN adaptation uses 32-dimensional embeddings, learned top-15 neighborhoods, graph attention and a shared two-layer head, with multiple channels and no batch normalization. Development-standardized errors average four trailing one-step predictions, then take a group maximum. It receives causal observed target history, an information advantage over the long-lead joint references. Its inherited group support and participation mapping are explicit adaptations, not a claim to reproduce the authors' published benchmark accuracy.

We report S (spatial entropy), PE and SE separately, T (both temporal entropies), ST (S plus T), spatial conventional SB, temporal conventional TB, B (SB plus TB), and BST (B augmented with S and T). Component families combine by maximum. A family ignores an unsupported component but abstains if none is available. Each level/change pair itself requires both endpoints. B versus BST tests incremental benefit rather than only replacement of conventional features.

D. Joint bootstrap and generative references

Both references generate joint time-by-sensor trajectories, conditioned on a 48-row context ending strictly before the 120-row target. Known calendar harmonics, coordinates and graph are allowed; future measurements are not. For a decision at t , issuance is $t - 120$. The $W = 96$ level uses forecast steps 25–120 and its previous endpoint steps 1–96. The $W = 48$ counterparts use steps 73–120 and 61–108. Thus decision lead is 240 minutes for Intel and 600 minutes for PEMS, a substantial forecasting challenge. Both feature endpoints come from the same generated trajectory.

The joint bootstrap chooses intact training blocks using historical-context and calendar similarity, sampling from the

nearest 64 donors. Donor masks remain. The diffusion model is the original compact DiffSTG/CSDI-inspired adaptation: width 24, three graph-temporal blocks, historical measurements/masks and fixed graph/calendar encodings. With Gaussian noise ϵ ,

$$Y_s = \sqrt{\bar{\alpha}_s} Y + \sqrt{1 - \bar{\alpha}_s} \epsilon, \quad (10)$$

$$\mathcal{L}_{\text{diff}} = \mathbb{E} \left[\frac{\|M_{\text{loss}} \odot (\epsilon - \epsilon_\theta(Y_s, s, C, G))\|_F^2}{\sum M_{\text{loss}}} \right]. \quad (11)$$

Only valid training targets enter the loss. Original checkpoints use a cosine schedule [16], 100 diffusion steps, 20-step DDIM [17], Adam 10^{-3} , batch 8, at most 2400 updates and development early stopping. Denoised states are clipped to $[-8, 8]$ training units; three training seeds 17/29/43 are retained. We do not tune a new architecture on inspected tests. Each reference supplies 64 draws. Observed masks are imposed after generation, and additional dropout masks are also applied to the reference measurements. Bootstrap donor masks can still reduce per-draw rows/templates; common-unit fidelity comparisons do not claim identical internal estimator support.

E. Scores, group aggregation and calibration

For feature a , set $u_a = (f_a, f_a - f_{a,t-d})$. Componentwise generated medians $c_{a,t,j}$ and MADs give

$$z_{a,t,j} = \frac{u_{a,t,j} - c_{a,t,j}}{1.4826 \text{ median}_b |u_{a,t,j}^{(b)} - c_{a,t,j}| + \epsilon_j}, \quad (12)$$

$$A_a(t) = \max_j |z_{a,t,j}|. \quad (13)$$

Floors for level/change features are $\max(10^{-4}, .05 \text{ IQR}_{dev})$. Legacy matrix-distance and raw-residual comparators retain .01 and .05 regularization floors, respectively. Reference-support counts are retained; the temporal rule requires at least 52 finite draws, while the retained spatial summary implementation uses 51 of 64. MAD scaling does not imply Gaussian tails. Actual Δf remains separate from $\Delta f - c_{\Delta f}$.

For each temporal group, average its largest $k = \lceil \eta m \rceil$ valid sensor scores, requiring at least $\lceil m/2 \rceil$ valid members. Fractions $\eta \in \{.25, .5\}$ and localization budgets in $\{6, 12, 24\}$ are selected jointly from four development copy/noise/drift/delay episodes. Mean localization IoU is pooled over T, TB, ST, B and BST and both references. All families use the same selected fraction and budget. Max-combined families receive their own calibration; no uncalibrated union of separately thresholded alerts is used.

With frozen family $\mathcal{F}$, let $M_t = \max_{a \in \mathcal{F}} A_a(t)$. Calibration units are separated by 168 rows (48 context plus 120 target), or are separate synthetic simulations. For n_c calibration maxima,

$$p_t = \frac{1 + \sum_{b=1}^{n_c} \mathbf{1}\{M_b \geq M_t\}}{n_c + 1}. \quad (14)$$

An entirely abstained unit has $M = -\infty$ and cannot alert. Primary $\alpha = .10$ is fixed before testing. Temporal dependence and distribution shift can invalidate exchangeability despite gaps; this is a per-issuance statement, not a stream-wide guarantee. Neither $1-p$ nor a model deviation is a physical-fault probability. Test controls evaluate background rates; separate

retrospective rate caps are explicitly descriptive, not prospective threshold validation.

F. Localization and availability

Group participation weights each group's score by inverse size and normalizes over eligible incident groups. Direct temporal ranking uses sensor scores; the combined rule averages their normalized rank percentiles. S/SB use participation, T/PE/SE/TB direct ranking, and ST/B/BST the combined rule. All three alternatives are also reported. The budget is selected on development data, never from test truth. Candidate-group best-overlap IoU is an oracle diagnostic, not a deployable method. Participation is not causal attribution.

Operational analysis includes every scheduled unit and injected event, including abstentions and misses. A separately calibrated common-support analysis uses group/window/channel units valid for S, PE, SE, SB and TB within each reference. Reference comparisons also report their differing model-support counts; common feature support is not claimed to eliminate every reference-support difference. Conditional localization is reported only alongside unconditional scores, where misses contribute zero. Quality flags, signed agreement and onset alignment are descriptive and do not create additional uncalibrated detections.

IV. THEORETICAL AND NUMERICAL CHARACTERIZATION

Proposition 1 (Bounds and shrinkage). *For a PSD correlation matrix, $0 \leq H_S \leq 1$. At $\lambda = 0$, identity has entropy one and rank one entropy zero. With positive shrinkage the lower bound is $-[a \log a + (m-1)b \log b] / \log m$, where $a = 1 - \lambda + \lambda/m$ and $b = \lambda/m$.*

Proof. The eigenvalues normalized by trace form a probability vector. Shannon entropy is nonnegative and maximized by uniform probabilities. Shrinkage maps its simplex into the convex hull of permutations of $(a, b, \dots, b)$; concavity makes these extreme points minimizers. $\square$

For equicorrelation $\mathbf{R}(\rho) = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$,

$$\frac{dH_S}{d\rho} = \frac{m-1}{m \log m} \log \frac{1-\rho}{1+(m-1)\rho} \leq 0, \quad 0 \leq \rho < 1. \quad (15)$$

The eigenvalues are $1 + (m-1)\rho$ and $1 - \rho$ with multiplicity $m-1$. Substitution proves the derivative; shrinkage replaces ρ by $(1-\lambda)\rho$ and multiplies by $1-\lambda$. The displayed nonpositive sign is restricted to nonnegative ρ , not the whole PSD interval.

Entropy need not be determined by average correlation. For $m = 4$, compare equicorrelation .2 with two independent two-sensor blocks of correlation .6. Both have $C = C_{abs} = .2$ and leading concentration .4, but unshrunk entropies are .960964 and .860964. Their full matrices distinguish them. This is an analytic example, not empirical evidence of detector superiority.

Proposition 2 (Order invariance on an aligned complete window). *For a common time-row permutation P , the correlation matrix and its correlation-spectrum entropy are unchanged.*

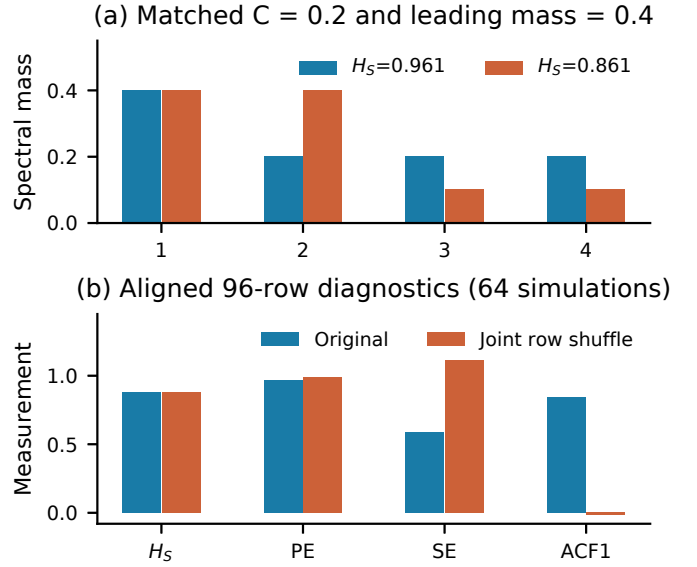


Fig. 2. Mathematical illustrations and aligned diagnostics, not primary fault performance. Equal average correlation and leading concentration can coexist with different spectral entropy. A joint time permutation preserves the evaluated matrix while temporal measurements can change.

Proof. Column centering and sample scaling commute with row permutation, and $(PZ)^\top(PZ)/(n-1) = Z^\top P^\top PZ/(n-1) = Z^\top Z/(n-1)$. $\square$

Order-sensitive temporal measurements can change. This elementary property applied to measurement design is not a new theorem. It does not imply invariance of every overlapping rolling window or the whole entropy-change trace. Offsets, nonzero column gains and sign conjugation also leave the correlation spectrum unchanged; temporal ordinal entropy can likewise miss amplitude changes.

Proposition 3 (Conditional rank calibration). *Conditional on a fixed scoring rule, if calibration and next-null maxima are exchangeable, (14) is super-uniform. Its smallest attainable value is $1/(n_c + 1)$.*

Proof. Among $n_c + 1$ exchangeable scores the test score's upper rank is uniform in the absence of ties; including all tied calibration scores makes it conservative. The trained rule must be fixed before these units. $\square$

The premise is not established merely by chronological partitioning. Controlled null simulations, finite-difference derivatives, explicit CPU pair counts, CPU/GPU agreement and leakage/event-matching checks accompany the proof.

V. EXPERIMENTAL DESIGN AND AUDIT

The protocol and configuration were committed before full extension runs (freeze 18b9c4a7); original revision b900da1b remains recoverable. The extension was motivated by inspected earlier findings, so reused-data comparisons are exploratory. Exactly three primary datasets are retained.

Synthetic sizes and graph configurations are not separate datasets.

Synthetic. The original random geometric graphs have 64, 128 or 256 nodes. Dynamics use $x_{t+1} = (.82I - .25L/\max_i \deg_i)x_t + Bf_t + .12\epsilon_t$, three spatially loaded OU drivers, calendar signals and burn-in 256. Spectral radius is .82. Training data and scaling are retained, but extension development (four 312-row simulations), calibration (32 of 256 rows) and test (four of 312 rows) use independent seeds $9000000 + 1000N + 100s + e$. Episodes never join at window boundaries. Synthetic display timestamps encode simulation phase.

Intel Lab. The original source [18] records 54 motes at roughly 31-second cadence, with Celsius, relative humidity, lux and voltage. We retain the original causal right-closed two-minute last-reading bins, masks and coordinate groups. Temperature and humidity are primary; other channels remain in the original auxiliary audit. The previously declared terminal-coverage rule ends quantitative data before March 24, 2004; 17,970 rows retain original 60/15/10/15 chronological partitions. The four extension test blocks were previously inspected. New injections do not make them wholly new validation.

PEMS-BAY. The DCRNN release [19] provides 325 road sensors, five-minute speed in mph over January–June 2017. The retained preprocessing expands 12 timestamp-gap rows, masks 521 zeros under release evaluation convention and retains published road distances. Per-value upstream imputation provenance is unavailable. The 52,128-row grid uses chronological 60/15/10/15 partitions. Extension blocks exclude original primary blocks and the original six-unit fidelity audit. A subsequent provenance audit found that the original long-window global diagnostic had nevertheless inspected every selected block. Both real-data extensions therefore reuse inspected backgrounds. These are unlabelled real backgrounds with semi-synthetic injections, not verified physical faults.

All normalizers use training medians and IQR/1.349 with documented fallbacks. Missing neural inputs use zero plus a mask; statistical estimators retain NaN. No observation is filled from a neighboring sensor. No window crosses a split. Real calibration partitions were already used in the original exploratory study; the extension’s new family thresholds are fitted there without test labels. Fitted models and groups are frozen. Calibration has 32 synthetic, 10 Intel and 31 PEMS issuances; minimum rank p-values are .0303, .0909 and .03125. Intel therefore has a particularly coarse operating point at nominal .10.

The heterogeneous benchmark retains 12 mechanisms: copying, common interference, decoupling, independent noise, delay, offset/gain drift, flatline, dropout, anticorrelated copy, mixed collapse/noise, matched-average-correlation covariance change and marginal-preserving temporal rearrangement. Each combines severity .6/1 with duration 12/48/96, giving 72 independently specified injection draws per configuration, sizes 6/12 and connected/partly disconnected groups. New fault seeds are $97100 + i$; seed $98100 + b$ defines controls. Synthetic coupling loss changes the state transition using the paired

clean simulation seed; real decoupling remains observational intervention. Onset is row 192; decisions are 167,179,...,311. The first in-event decision is 203, already 11 samples after onset. Untouched and legitimate synchronized-transition controls accompany each base. Fault trials therefore comprise 90% of this constructed evaluation; its precision is not a real-deployment base-rate estimate. Forecast contexts end before every injected onset.

An alert is a connected run of consecutive flagged decisions. Hungarian one-to-one matching credits only an alert whose onset lies inside the event, with no trailing tolerance or point adjustment. Unmatched early, late and control alarms are counted in controlled-benchmark precision. Native-control exceedances are background alerts, not certified false physical diagnoses. Event PR area uses the declared upper precision envelope over reachable recalls because alert merging can make event recall nonmonotone. Detection delay includes the decision grid; measured pipeline time is reported separately.

Paired percentile intervals resample whole simulations or disjoint recording blocks, retaining their repeated injections and three model seeds. Overlapping windows are not independent replicates. Four blocks per configuration give limited real-data precision, and adjacent recording blocks need not be independent. Intervals are exploratory and unadjusted for multiple comparisons; there is no predeclared equivalence test. We report mechanism, severity, duration, direction and support strata, rather than infer equivalence from failure to reject a difference.

Two scoring distinctions are documented in the audit. Original finite-component maxima could use a level when its change was unavailable; the extension requires both endpoints of each pair. The original matrix statistic compared current $\mathbf{R}$ with its reference, whereas the extension also measures $\Delta\mathbf{R}$. Original results are not silently relabeled as corrected extension results. Original removed/shuffled-graph ablations remain in v1; the temporal extension fixes the physical/proximity graph for its feature comparisons. An interrupted integration attempt is included in the runtime ledger. Pure participation batching was checked against the original weighted formula.

VI. RESULTS

A. Primary comparisons and incremental benefit

Table I reports all scheduled events; Fig. 3 adds paired block intervals. RQ1 has a mixed answer. Synthetic diffusion S has lower recall than SB by -0.099 [-0.174, -0.034]; its IoU difference is -0.010 [-0.017, -0.003]. Intel diffusion S/SB recall is 0.213/0.481, but their background exceedances are 0.218/0.372. PEMS bootstrap S appears stronger than SB at nominal alpha (0.528 versus 0.005 recall), with background rates 0.301 versus 0.000. Under the descriptive test-control cap .10, corresponding recalls become 0.069 and 0.264, at attainable rates 0.026 and 0.064. Thus a nominal-threshold ranking is not an isolated test of feature information. The retrospective analysis does not validate a deployable threshold.

For RQ3, synthetic bootstrap T/TB recall is 0.401/0.532, with paired difference -0.131 [-0.313, +0.054]; unconditional IoU is

0.093/0.239. Intel bootstrap T instead exceeds TB by +0.389 [+0.264, +0.523] recall, but their achieved background rates differ. PE and SE are not interchangeable: Intel bootstrap PE/SE recall is 0.310/0.380, with IoU 0.022/0.073. Switching Intel T to diffusion yields recall 0.060 and background exceedance 0.288. PEMS temporal contrasts have wide block intervals. These results do not establish a universal entropy ordering.

For RQ4, synthetic bootstrap ST improves on S by +0.123 [+0.028, +0.230] recall, but ST minus T is +0.008 [-0.085, +0.127]. Their primary localization rules differ; fixed-rule alternatives are retained to distinguish feature and ranking effects. The direct incremental comparison BST minus B is +0.003 [+0.000, +0.008] recall for synthetic bootstrap. The other five dataset/reference combinations have zero mean recall change at this operating point. This is not equivalence: different feature information can be suppressed by a maximum and its recalibrated threshold. Synthetic bootstrap BST minus B IoU is +0.008 [+0.000, +0.017]. Both use the same combined localization rule. A saved maximum-dominance audit shows that conventional components usually determine the final scan maximum. The retained causal GDN adaptation has synthetic/Intel/PEMS recall 0.514/0.139/0.157, with background rates 0.135/0.436/0.038. PEMS bootstrap raw residuals reach 0.264 recall at 0.077 background exceedance. These additional detectors retain their disclosed support and information-access differences. For bootstrap T, conditional versus unconditional IoU is 0.231/0.093 on synthetic, 0.195/0.076 on Intel and 0.038/0.015 on PEMS. Mean best-overlap candidate-group IoU is 0.284/0.413/0.325, an oracle library diagnostic rather than an upper bound on every sensor ranking.

B. Information captured and mechanism dependence

RQ2 is affirmative as an information distinction, not a superiority claim. In 64 aligned simulations, joint row permutation changes the evaluated correlation matrix by at most 1.3×10^{-15} (rounded upward). Mean PE changes from 0.964 to 0.989, SE from 0.586 to 1.115, and ACF1 from 0.845 to -0.016; spatial entropy is unchanged. Conversely, copying lowers mean spatial entropy from 0.877 to 0.134, while mean PE changes only from 0.964 to 0.965. Offsets, gains, sign changes, quantization, replay, periodic interference, irregularity, legitimate transitions and dropout remain separate diagnostic cases. *These deliberately constructed experiments are not the primary benchmark.*

In the heterogeneous synthetic benchmark, bootstrap T/TB recall on flatlining is 0.741/0.241; for independent noise it is 0.463/1.000. These are mechanism-specific operational results, not calibrated guarantees at identical false-alert rates. A rolling window crossing a flatline onset can remain measurable; a wholly constant window correctly abstains. The artifact reports paired mechanism, severity, duration, measured-direction and affected-support strata. Direction is the measured lagged change, not a fault-name label or deviation from reference expectation. The first scheduled in-event decision is already 11 samples late. Short events can therefore be missed before adequate window evidence becomes available.

C. Reference fidelity, support and calibration

For RQ5, the shared-support audit in Table II distinguishes reference defects from feature limits. Synthetic64 diffusion spatial-entropy coverage is 0.453 versus 0.740 for bootstrap; Intel values are 0.083 and 1.000, on only 6.000 common spatial units per audit issuance on average. Intel diffusion has better raw energy score (6.336 versus 7.234), yet its sample-entropy coverage is 0.000 versus 0.953 for bootstrap. Good raw marginal behavior does not ensure a faithful temporal functional. No consistent generative-reference improvement is established, and this compact long-lead model does not represent all generative approaches.

RQ6 exposes distinct operational and statistical limitations (Fig. 5). Intel S group/window eligibility is 0.017 under bootstrap and 0.049 under diffusion; T eligibility is 0.541 and 0.633. Per-sensor support avoids simultaneous complete rows across a group, but availability alone does not guarantee calibrated detection. Recalibrating on common feature support can also change achieved rates: Intel bootstrap common-support B background exceedance is 0.551, compared with 0.000 operationally. Varying support and nonexchangeable calibration/test distributions remain unresolved by nominal rank calibration.

Development selects $q = 3$ throughout, delay 1 for synthetic and 2 for both real datasets; $q = 4$ loses support at these short windows. Sample tolerance and group fraction are saved per configuration. At W96, scale 2 can retain sufficient templates, whereas scale 4 cannot meet the declared 30-template minimum; W24 is unsupported for the primary temporal entropies. We report these as limited multiscale support diagnostics; a separately frozen scale-2 detector sensitivity is reported below.

The seed-17, separately calibrated W48/W96 and 51-versus-52-reference-draw support sensitivity completed 5 of five configurations. Across completed configurations, the largest absolute recall change caused by the support-rounding correction is 0.000. Window-specific thresholds and results are retained separately; the 120-step forecast lead is held fixed.

The separately frozen, exploratory seed-17 scale-2 comparison keeps W96's physical history, $d=24$ and both references. All temporal measurements use the same two-row averages; spatial measurements remain on the original grid. Bootstrap T recall changes (scale 2 minus 1, paired block intervals) are -0.093 [-0.231, +0.069], -0.069 [-0.458, +0.333] and -0.333 [-0.722, -0.028] for synthetic, Intel and PEMS. Corresponding TB changes are -0.074 [-0.245, +0.042], +0.264 [+0.000, +0.750] and -0.153 [-0.278, -0.028]. Full PE/SE, localization, support and achieved background rates accompany these separately calibrated comparisons in the artifact. Scale-1 calibration ranks reproduce the independent W96 audit; S/SB scores are unchanged across scales. This reused-data sensitivity does not select a new primary scale.

A post-hoc CPU estimator check uses 128 independent unit-variance Gaussian AR(1) realizations. For independent continuous observations, PE's between-realization SD is 0.023 at W48 and 0.009 at W96. With fixed sample-entropy tolerance

TABLE I

OPERATIONAL EXTENSION RESULTS AT NOMINAL $\alpha = .10$. P/R: EVENT PRECISION/RECALL; AP: EVENT-PR ENVELOPE AREA; IOU INCLUDES MISSED EVENTS AS ZERO. THESE ARE NOT MATCHED ACHIEVED BACKGROUND RATES. PE/SE, ALL COMPARATORS AND COMMON-SUPPORT RESULTS ARE RETAINED IN THE ARTIFACT.

Reference	Family	Synthetic (three sizes)				Intel Lab				PEMS-BAY			
		P	R	AP	IoU	P	R	AP	IoU	P	R	AP	IoU
Block	S	0.256	0.285	0.220	0.024	0.251	0.208	0.083	0.017	0.211	0.528	0.179	0.035
Block	T	0.329	0.401	0.290	0.093	0.329	0.389	0.172	0.076	0.291	0.389	0.229	0.015
Block	ST	0.308	0.409	0.272	0.075	0.329	0.389	0.172	0.070	0.291	0.389	0.229	0.018
Block	B	0.484	0.650	0.452	0.193	0.000	0.000	0.139	0.000	0.224	0.148	0.207	0.030
Block	BST	0.486	0.653	0.456	0.200	0.000	0.000	0.139	0.000	0.224	0.148	0.207	0.030
Diffusion	S	0.197	0.170	0.196	0.010	0.158	0.213	0.086	0.014	0.373	0.287	0.281	0.006
Diffusion	T	0.227	0.318	0.209	0.044	0.048	0.060	0.051	0.002	0.276	0.403	0.225	0.011
Diffusion	ST	0.197	0.170	0.196	0.009	0.048	0.060	0.051	0.002	0.376	0.287	0.300	0.001
Diffusion	B	0.327	0.392	0.284	0.102	0.444	0.056	0.120	0.015	0.267	0.111	0.211	0.022
Diffusion	BST	0.327	0.392	0.284	0.103	0.444	0.056	0.120	0.011	0.267	0.111	0.211	0.022

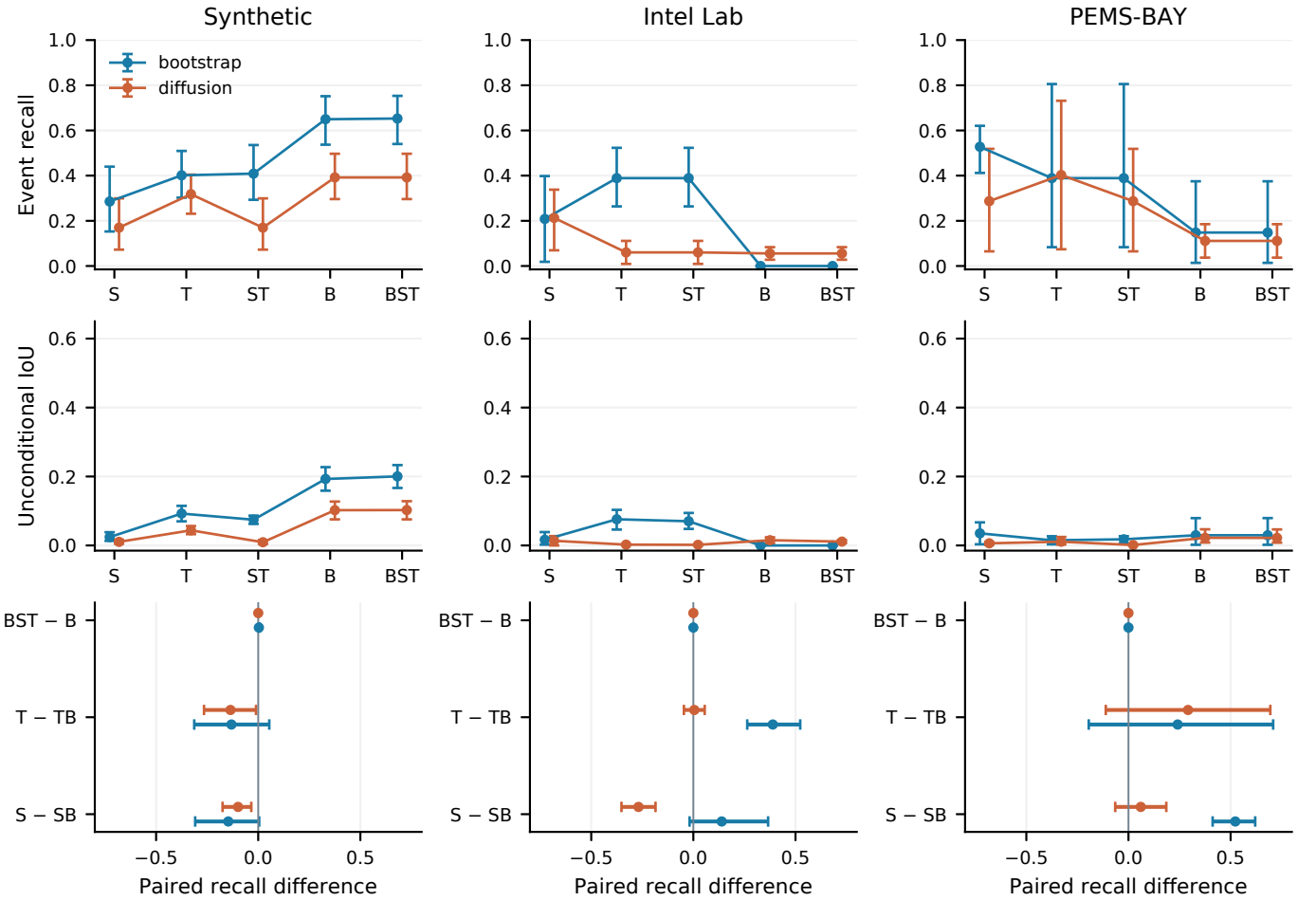


Fig. 3. Operational performance and unadjusted paired 95% whole-block intervals. IoU assigns zero to missed events. Equal nominal alpha does not imply equal achieved background rates; four real recording blocks give limited precision.

.2, finite supported fractions are 0.086 and 1.000; conditional means therefore require caution. At $\rho = .8$, quantization to .5-unit steps yields tied-template fraction 0.653. Stable-tie PE eligibility is 0.008, versus 0.758 after discarding ties. This demonstrates sensitivity of the quality/support decision, not

a reason to reinterpret arbitrary tied ordering as confident dynamics.

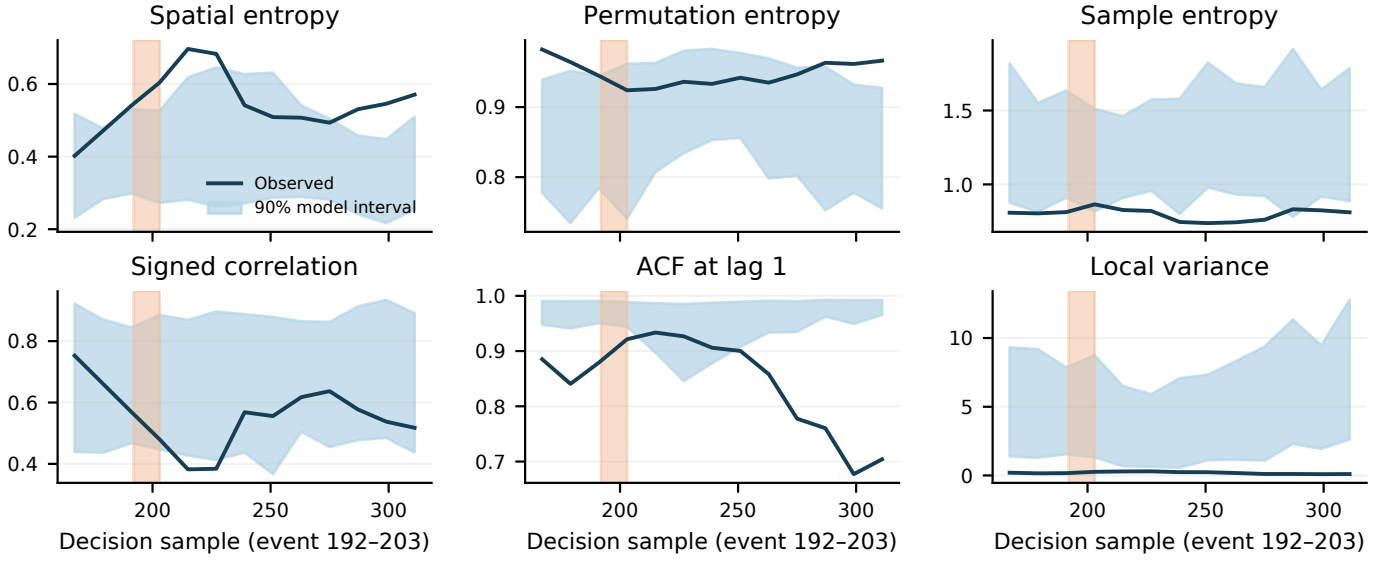


Fig. 4. First prespecified synthetic copy episode: common forecast draws supply all bands. The first affected sensor and the best-overlap candidate group are chosen for illustration only, using injection labels; they are not a detection/localization rule. Shading marks the 12-row injection.

TABLE II

PAIRED FIDELITY ON UNCHANGED BACKGROUNDS: SEED 17, FIRST TWO BLOCKS, DECISIONS 167/263. REFERENCES SHARE EVALUATED RAW COORDINATES OR ELIGIBLE FEATURE UNITS; DONOR MASKS CAN STILL CHANGE PER-DRAW SUPPORT. RAW, H, PE, SE: 90% COVERAGE; TV: ORDINAL-PATTERN TOTAL VARIATION; ES: NORMALIZED ENERGY SCORE (LOWER IS BETTER). TEMPORAL ENTRIES USE $W=96$. MEANS OMIT UNDEFINED UNITS; COUNTS AND WIDTHS ARE SAVED.

Dataset	Reference	Raw	H	PE	SE	ACF1 MAE	Ordinal TV	R RMSE	ES
Synthetic	Block	0.875	0.812	0.855	0.858	0.034	0.081	0.236	0.578
Synthetic	Diffusion	0.803	0.513	0.429	0.222	0.047	0.150	0.313	1.747
Intel Lab	Block	0.616	1.000	0.887	0.953	0.013	0.378	0.467	7.234
Intel Lab	Diffusion	0.714	0.083	0.214	0.000	0.114	0.406	0.525	6.336
PEMS-BAY	Block	0.847	0.819	0.873	0.862	0.121	0.094	0.288	1.501
PEMS-BAY	Diffusion	0.739	0.494	0.699	0.430	0.163	0.118	0.360	2.055

VII. DASHBOARD AND COMPUTATIONAL ARTIFACT

The dashboard keeps node coordinates fixed and distinguishes physical edges from measured correlation-departure overlays. It links candidate groups to per-sensor PE/SE, conventional temporal traces, predictive bands, actual changes, reference departures and valid-template/pair counts. Shared channels and signed agreement are measured descriptions. Onset-alignment summaries do not add a decision rule. Forecast issuance, observation time, checkpoint version and scan-calibrated p-value remain visible. Three prespecified first-copy examples are replayed; they are not selected to maximize apparent detection success.

CUDA batches covariance, eigendecomposition, ordinal counting, conventional statistics and joint generation. Sample-entropy pairs are chunked (512 pairs, 1024 series) to control quadratic intermediates; statistical kernels remain float32, checked against CPU float64. Neural mixed precision is separate. The artifact retains configuration/split/model hashes, exact prediction and calibration arrays, per-sensor support, compact replay and all joint reference trajectories on the experiment host. Bulky reproducible arrays are excluded

from Git; manifests and reconstruction commands describe them. Kernel timings after warmup are distinguished from complete-pipeline latency. Isolated generation-to-score medians for synthetic64, Intel and PEMS are 0.180, 0.174 and 0.258 seconds for bootstrap, versus 0.370, 0.338 and 1.164 for diffusion on the RTX A6000. They include generation, masks, both windows, feature extraction, scoring and localization; they exclude GDN, acquisition, loading, disk output, rank lookup and dashboard rendering. They are not a production end-to-end speedup claim. Peak allocated memory in these isolated runs is at most 1.43 GiB. The PEMS temporal kernel takes 0.724 seconds on CPU float64 versus 0.035 on GPU float32, transfer excluded; precision and hardware both differ. Maximum CPU/GPU absolute disagreement across the benchmark is below 2.0×10^{-7} . The cumulative GPU-stage ledger, including the original study, validation and failed attempts, records 4.271 hours within the explicitly authorized eight-hour ceiling. All primary, window/support, scale-2 and fidelity/timing comparisons completed. The earlier four-hour stop is preserved; CPU reporting and rendering are outside this experiment-stage ledger.

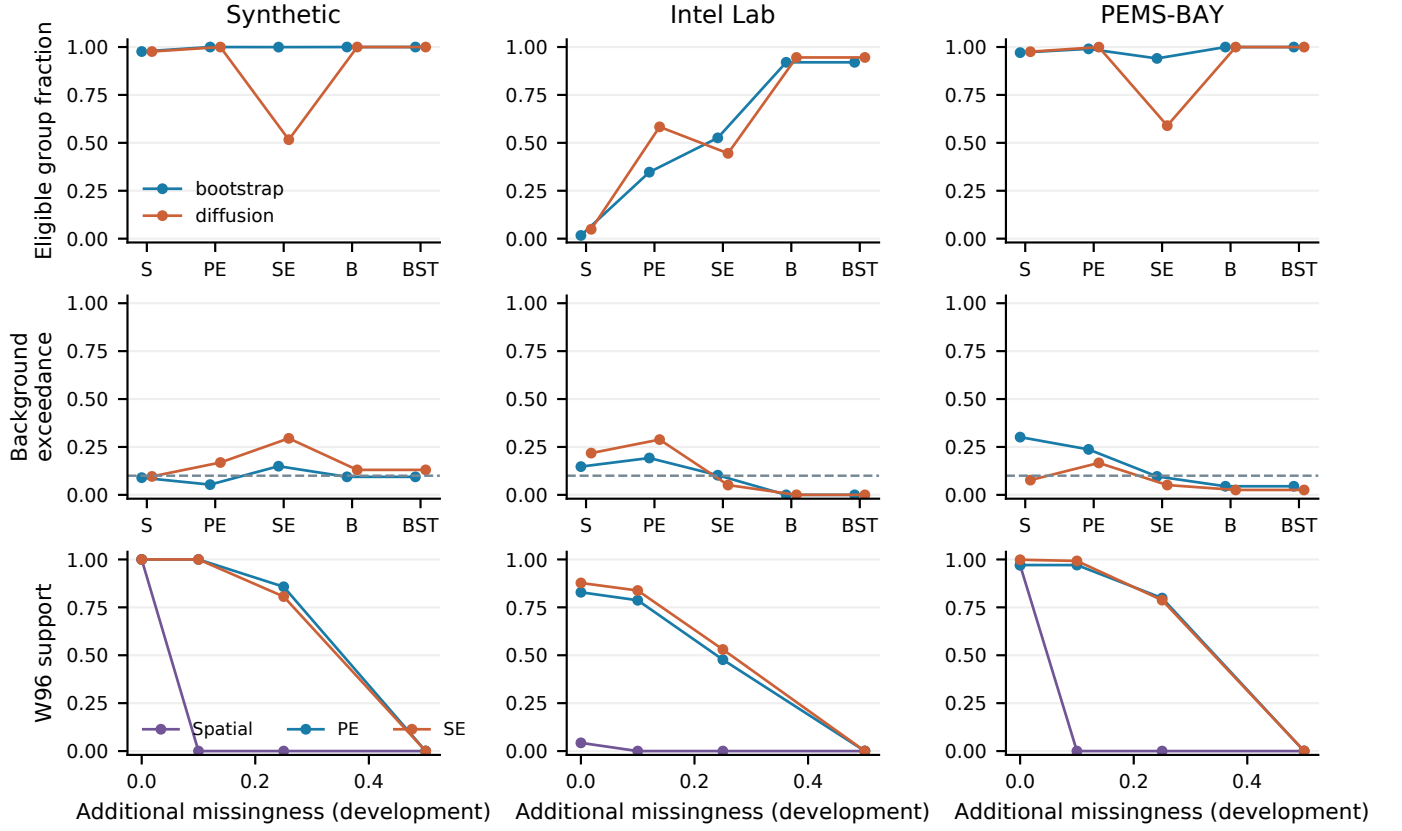


Fig. 5. Top: operational eligible-group fraction. Middle: untouched-background exceedance at nominal α .10. Bottom: development-only support after independent added missingness, W96; spatial is group support, PE/SE per-sensor support. These different denominators are intentional and explicit.

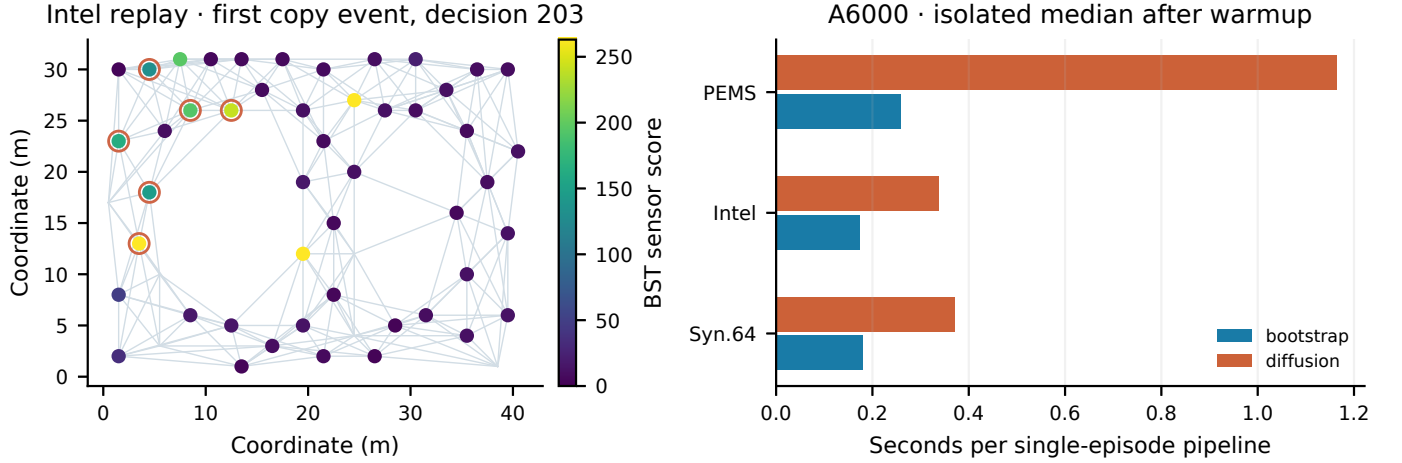


Fig. 6. Intel replay localization display (orange outlines: injected sensors, for evaluation only) and isolated generation-to-score timing. The example is the first prespecified copy trial, not a selected success. Timing excludes loading, acquisition and presentation.

VIII. LIMITATIONS AND CONCLUSION

The comparison establishes different measurement information, but no consistent overall entropy advantage. Temporal entropy can respond when aligned covariance is invariant; conventional temporal statistics can respond too. Missing-data support and reference mismatch materially affect operational results. The Intel bootstrap temporal-entropy advantage occurs

under one frozen calibration setting with different achieved background rates, while synthetic conventional temporal localization is stronger. Augmenting conventional features changes synthetic bootstrap recall by 0.003 and leaves recall unchanged in the other primary dataset/reference combinations. This limited incremental effect under a maximum rule does not establish statistical or practical equivalence. The evaluated

diffusion reference shows no consistent improvement over joint bootstrap; its feature-distribution failures cannot establish that entropy itself is intrinsically inadequate. The real-data study is semi-synthetic and partly reuses inspected recordings; it cannot establish sensitivity to verified native faults. Compact diffusion forecasts at long lead, model-specific support, limited calibration units, four real recording blocks and a finite neighborhood library constrain the comparison. Unsupported sample entropy is retained as a limitation, not silently replaced. Longer multiscale histories would need equally long conventional comparators and new forecasting/calibration evaluation. Native causes and dashboard usability are not validated by injected labels.

ACKNOWLEDGMENT AND AI DISCLOSURE

OpenAI Codex assisted implementation, experiment orchestration, analysis code, figure generation and drafting throughout this work. Numerical results come from the saved experiments and stated proofs; the authors retain responsibility for verification and final submission. The diffusion reference is a separate scientific model described in the methods. We acknowledge the original Intel Lab data collectors and the DCRNN release authors.

REFERENCES

- [1] A. V. Mantzaris, Y.-H. Chen, G.-R. Domenikos, and L. Choudur, "Exploring the effects of urban network topologies on entropy trajectories of segregation," *Scientific Reports*, vol. 14, no. 1, 2024, art. no. 19188.
- [2] A. V. Mantzaris and G.-R. Domenikos, "Exploring the thermodynamic description of a simulation of flocking birds," *Frontiers in Complex Systems*, vol. 2, 2025, art. no. 1516812.
- [3] O. Roy and M. Vetterli, "The effective rank: A measure of effective dimensionality," in *European Signal Processing Conference*, 2007. [Online]. Available: <https://zenodo.org/records/40328>
- [4] P.-Y. Chen, L. Wu, S. Liu, and I. Rajapakse, "Fast incremental von Neumann graph entropy computation: Theory, algorithm, and applications," in *International Conference on Machine Learning*, vol. 97, 2019, pp. 1091–1101.
- [5] C. Bandt and B. Pompe, "Permutation entropy: A natural complexity measure for time series," *Physical Review Letters*, vol. 88, no. 17, p. 174102, 2002.
- [6] J. S. Richman and J. R. Moorman, "Physiological time-series analysis using approximate entropy and sample entropy," *American Journal of Physiology–Heart and Circulatory Physiology*, vol. 278, no. 6, pp. H2039–H2049, 2000.
- [7] M. Costa, A. L. Goldberger, and C.-K. Peng, "Multiscale entropy analysis of complex physiologic time series," *Physical Review Letters*, vol. 89, no. 6, p. 068102, 2002.
- [8] K. Rasul, C. Seward, I. Schuster, and R. Vollgraf, "Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting," in *International Conference on Machine Learning*, vol. 139, 2021, pp. 8857–8868.
- [9] Y. Tashiro, J. Song, Y. Song, and S. Ermon, "CSDI: Conditional score-based diffusion models for probabilistic time series imputation," in *Advances in Neural Information Processing Systems*, vol. 34, 2021. [Online]. Available: https://proceedings.nips.cc/paper_files/paper/2021/hash/cfe8504bda37b575c70ee1a8276f3486-Abstract.html
- [10] H. Wen, Y. Lin, Y. Xia, H. Wan, Q. Wen, R. Zimmermann, and Y. Liang, "DiffSTG: Probabilistic spatio-temporal graph forecasting with denoising diffusion models," in *Proceedings of the 31st ACM International Conference on Advances in Geographic Information Systems*, 2023, pp. 1–12.
- [11] C. Xiao, Z. Gou, W. Tai, K. Zhang, and F. Zhou, "Imputation-based time-series anomaly detection with conditional weight-incremental diffusion models," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 2742–2751.
- [12] Y. Chen, C. Zhang, M. Ma, Y. Liu, R. Ding, B. Li, S. He, S. Rajmohan, Q. Lin, and D. Zhang, "ImDiffusion: Imputed diffusion models for multivariate time series anomaly detection," *Proceedings of the VLDB Endowment*, vol. 17, no. 3, pp. 359–372, 2023.
- [13] F. Zhang, S. Chin, J.-H. Xue, and W. Yang, "ICDiffAD: Implicit conditioning diffusion model for time series anomaly detection," in *International Conference on Learning Representations*, 2026.
- [14] A. Deng and B. Hooi, "Graph neural network-based anomaly detection in multivariate time series," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 5, pp. 4027–4035, 2021.
- [15] V. Chernozhukov, K. Wüthrich, and Y. Zhu, "Exact and robust conformal inference methods for predictive machine learning with dependent data," in *Conference on Learning Theory*, vol. 75, 2018, pp. 732–749.
- [16] A. Q. Nichol and P. Dhariwal, "Improved denoising diffusion probabilistic models," in *International Conference on Machine Learning*, vol. 139, 2021, pp. 8162–8171.
- [17] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *International Conference on Learning Representations*, 2021. [Online]. Available: <https://arxiv.org/abs/2010.02502>
- [18] P. Bodik, W. Hong, C. Guestrin, S. Madden, M. Paskin, and R. Thibaux, "Intel lab data," 2004, accessed September 30, 2026. [Online]. Available: <https://db.csail.mit.edu/labdata/labdata.html>
- [19] Y. Li, R. Yu, C. Shahabi, and Y. Liu, "Diffusion convolutional recurrent neural network: Data-driven traffic forecasting," in *International Conference on Learning Representations*, 2018. [Online]. Available: <https://arxiv.org/abs/1707.01926>